Correlation-Guided Adaptive Penalized SBL for Joint Activity Detection and Channel Estimation in MTC

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***Abstract*—Machine-type communications (MTC) play a crucial role in modern wireless networks, particularly in Internet of Things (IoT) applications involving massive numbers of machine-type devices (MTDs). The event-driven nature of MTC can produce spatially correlated and bursty traffic, presenting challenges for joint activity detection and channel estimation (JADCE) in grant-free access systems. While sparse Bayesian learning (SBL) exploits the sparsity of MTD activity to enable recovery using non-orthogonal pilot sequences, conventional SBL does not explicitly incorporate spatial dependence among MTD activity states. To address this limitation, this paper proposes a correlation-guided adaptive penalized sparse Bayesian learning (CGAP-SBL) framework that incorporates spatial information into the hyperparameter estimation process. A Gaussian-inspired quadratic interaction model is employed to construct iteration-dependent, non-negative penalty coefficients that adaptively influence the SBL hyperparameter updates. A closed-form update is derived by optimizing the resulting penalized expectation-maximization (EM) subproblem. Furthermore, an adjusted negative correlation (ANC) approach is developed to construct the spatial interaction matrix, resulting in the CGAP-SBL-ANC algorithm. The proposed approach is evaluated through comparisons with conventional sparse recovery methods under event-driven spatially correlated activity. A separate spatial-randomization experiment is designed to investigate the influence of spatial activity dependence on detection and channel estimation performance while preserving the realized number of active devices.**

***Index Terms*—Adaptive penalization, Channel estimation, Correlated device activity, Machine-type communications, Sparse Bayesian learning.**

I. Introduction

THE increasing demand for enhanced connectivity has driven the evolution of wireless communications from human-type communications (HTC) toward machine-type communications (MTC). Unlike HTC, which primarily facilitates human interactions, MTC enables autonomous interactions among machine-type devices (MTDs) without human intervention [1]. With MTC, machines can autonomously communicate wirelessly with a central node, forming networks of connected devices commonly associated with the Internet of Things (IoT) [2].

Unlike HTC, MTC traffic often exhibits bursty activity patterns, particularly in event-triggered applications. For instance, when smoke detectors trigger transmissions in an MTC network to signal a potential fire, nearby MTDs may become active simultaneously. Such scenarios can result in spatially correlated traffic among MTDs during critical events [3]. Additionally, MTC traffic is predominantly uplink-oriented, meaning that event-driven bursts of communication can place significant demands on receiving infrastructure, such as base stations (BSs) [4]. Despite the potentially high number of simultaneous transmissions, the messages exchanged in MTC are typically short. Since the signaling overhead required for conventional data transmission can be substantial relative to the payload size, efficient resource management becomes particularly challenging [5].

MTC networks can support significantly larger numbers of devices per cell than conventional HTC systems, with many devices exhibiting sporadic uplink activity. Consequently, conventional grant-based transmission schemes designed for HTC may be inefficient for MTC in terms of energy consumption and signaling overhead [4], [6].

Grant-free access protocols have emerged as a key enabler for MTC, offering energy-efficient and low-latency channel access. More precisely, grant-free access allows MTDs to transmit without requiring a separate scheduling grant for each transmission. Furthermore, owing to the high density of connected MTDs and the limited pilot resources, grant-free access systems commonly employ non-orthogonal pilot sequences. While these sequences reduce pilot resource requirements, they introduce a significant challenge: determining which MTDs are active while simultaneously estimating their channel state information (CSI). This problem is commonly known as joint activity detection and channel estimation (JADCE). The sporadic and sparse nature of MTC activity motivates the use of compressive sensing (CS) techniques to address the JADCE problem [7].

Traffic modeling plays a vital role in analyzing MTC networks. It is generally classified into two categories: source traffic modeling and aggregated traffic modeling. The primary distinction is that aggregated traffic modeling treats the traffic generated by multiple MTDs collectively, whereas source traffic modeling represents the traffic characteristics of individual devices. In aggregated models, traffic is characterized through aggregate statistical properties, while source models can assign device-specific parameters to describe individual traffic behavior. Although source traffic models provide greater flexibility in representing individual device characteristics, they generally require more detailed information and computational resources [4], [8]. For event-driven MTC, representing the spatial relationships among individual MTDs is particularly relevant because neighboring devices may respond to common triggering events.

Sparsity is a key characteristic of MTC traffic, as only a small subset of MTDs is typically active at a given time. This property can be exploited to identify active devices

and estimate their channels using limited pilot resources. In conventional orthogonal pilot-based access, the required pilot resources increase with the number of devices. In contrast, grant-free MTC systems can employ shorter non-orthogonal pilot sequences by exploiting sparse device activity. However, interference among non-orthogonal pilot sequences complicates the identification of active MTDs and the estimation of their channels. To address this challenge, multi-user detection (MUD) and CS techniques have been investigated [9]. Various CS-based methods have been proposed for active MTD detection and sparse channel recovery, including greedy algorithms, iterative techniques, convex optimization approaches, and sparse Bayesian learning (SBL) [10].

SBL provides a probabilistic framework for sparse signal recovery by modeling the unknown signal coefficients using parameterized prior distributions. Unlike approaches that rely exclusively on deterministic sparsity penalties, SBL estimates the parameters controlling these prior distributions from the received observations. In the conventional formulation, sparsity is promoted by learning continuous variance hyperparameters through type-II maximum likelihood estimation. This allows irrelevant signal components to be progressively suppressed while retaining a posterior representation of the remaining components. SBL has therefore been widely investigated for sparse signal recovery and JADCE in wireless communication systems [10]–[12]. However, conventional SBL does not explicitly exploit spatial dependence among device activity states, which can provide additional structural information in event-driven MTC networks.

A. Related Work

Sparse signal recovery techniques have become essential in JADCE for grant-free MTC, where sporadic transmissions and limited pilot resources challenge conventional access schemes. CS and SBL approaches exploit the sparsity of MTD activity to improve device detection and channel estimation performance.

Greedy algorithms such as orthogonal matching pursuit (OMP) [13] and compressive sampling matching pursuit (CoSaMP) [14] offer computationally efficient approaches to sparse recovery under limited pilot resources. Approximate message passing (AMP) algorithms are attractive for large-scale systems and are particularly effective under independent and identically distributed (i.i.d.) sensing matrix assumptions [15]. AMP has also been extended to support asynchronous access through successive interference cancellation (SIC) [16]. Generalized approximate message passing (GAMP) extends AMP by accommodating more general output channel models and input prior distributions [17]. By incorporating expectation-maximization (EM), EM-AMP-based methods can further exploit unknown statistical parameters in sparse recovery problems [18]. A Bayesian AMP approach has also been proposed for JADCE in spatially correlated multiple-input multiple-output (MIMO) channels by exploiting channel distribution information (CDI) [19]. Moreover, JADCE has been formulated using mixed-norm minimization [20], [21] and maximum a posteriori (MAP) estimation [22], with optimization approaches based on the alternating direction method of multipliers (ADMM).

Various forms of SBL algorithms have been developed for general sparse signal recovery. The original SBL framework was introduced for systems with a single observation vector [23], while multiple sparse Bayesian learning (M-SBL) extended the formulation to multiple measurement vector (MMV) settings, enabling joint recovery of sparse signals sharing a common support [24]. Several variants have subsequently been developed to improve computational efficiency, including covariance-free expectation-maximization (CoFEM) [11]. GAMP-SBL hybrids [25] and inverse-free Bayesian updates [26] have further enhanced the applicability of SBL to large-scale sparse recovery problems.

SBL has received considerable attention for JADCE in MTC owing to its ability to incorporate statistical information and exploit structured sparsity. It has been applied to sparse channel estimation in millimeter-wave (mmWave) MIMO systems [27] and JADCE in grant-free low-earth orbit (LEO) satellite networks [28]. To exploit channel correlation in massive MIMO systems, block sparse Bayesian learning (BSBL) [29] and its low-complexity variant AMP-BSBL [30] have been investigated, while GAMP-SBL has been applied in non-orthogonal multiple access (NOMA) systems [31]. Adaptive expectation-maximization SBL (AEM-SBL) has also been investigated for recovery under clustered sparsity patterns [32]. Other extensions have explored structured Bayesian priors and temporal activity dependence in grant-free MTC [33], [34]. Additionally, structured multiuser detectors based on M-SBL and BSBL, including MSBL-MUD and BSBL-MUD, have been proposed for uplink grant-free NOMA with variable user activity [35]. Regularization-based extensions, including total variation SBL (TV-SBL), have also been investigated to exploit dependencies among sparsity-controlling hyperparameters [36].

Although these approaches demonstrate the benefits of statistical structure in sparse recovery, spatial correlation in wireless channel coefficients and spatial correlation in MTD activity states represent different forms of dependence. In event-driven MTC, nearby MTDs may exhibit correlated activation patterns even when their physical channel realizations are modeled independently. Consequently, there is motivation for SBL approaches that explicitly incorporate modeled spatial activity relationships into the estimation of device-specific sparsity hyperparameters. In particular, the construction of spatial interaction matrices and their integration into computationally tractable adaptive hyperparameter updates remain important considerations for JADCE under event-driven traffic.

B. Main Contribution

This work addresses the JADCE problem in grant-free MTC systems by proposing a correlation-guided adaptive penalized sparse Bayesian learning (CGAP-SBL) framework that incorporates spatially correlated MTD activity into the hyperparameter estimation process. An adjusted negative correlation (ANC) approach is further developed as a spatial interaction matrix construction for the proposed framework, resulting in

the CGAP-SBL-ANC algorithm. The main contributions are summarized as follows:

- **Event-driven spatial activity modeling:** The JADCE problem is considered under an event-driven activity model in which MTD activations exhibit spatial dependence due to common triggering events. The model captures distance-dependent activation behavior and provides a basis for evaluating sparse recovery methods under spatially correlated MTD traffic.
- **CGAP-SBL framework:** A correlation-guided adaptive penalized SBL framework is developed by introducing a Gaussian-inspired quadratic interaction model over continuous channel-variance hyperparameters. The interaction gradient is used to construct iteration-dependent, non-negative penalty coefficients that incorporate spatial information into the SBL hyperparameter estimation process. The framework retains the conventional complex-MMV posterior estimation procedure while modifying the hyperparameter update through adaptive spatial penalization.
- **Analytical derivation and ANC construction:** A closed-form hyperparameter update is derived by maximizing an iteration-specific penalized EM auxiliary function, and its mathematical properties are analyzed. Furthermore, the ANC approach is developed to construct a spatial interaction matrix from modeled spatial relationships among MTDs. Integrating ANC with the proposed CGAP-SBL framework yields the CGAP-SBL-ANC algorithm.
- **Performance evaluation:** Numerical simulations are designed to evaluate the activity detection accuracy, channel estimation performance, and convergence behavior of CGAP-SBL-ANC against conventional sparse recovery benchmarks under event-driven spatially correlated activity. A controlled spatial-randomization experiment is additionally considered to assess the contribution of spatial activity dependence.

C. Notation

Throughout this paper, scalars, vectors, and matrices are denoted by italic letters, bold lowercase letters, and bold uppercase letters, respectively. The sets of real, non-negative real, strictly positive real, and complex numbers are denoted by $\mathbb{R}$, $\mathbb{R}_+$, $\mathbb{R}_{++}$, and $\mathbb{C}$, respectively. The operators $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^{-1}$ denote transpose, Hermitian transpose, and matrix inverse, respectively. The operator $\mathrm{diag}(\cdot)$ constructs a diagonal matrix, while $\mathbf{I}_n$ and $\mathbf{1}_{n\times n}$ denote the $n \times n$ identity and all-ones matrices, respectively. The Euclidean and Frobenius norms are denoted by $\|\cdot\|_2$ and $\|\cdot\|_F$, respectively. The operators $\mathrm{Tr}(\cdot)$ and $\det(\cdot)$ denote the matrix trace and determinant, respectively. The expectation operator is denoted by $\mathbb{E}[\cdot]$, and $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a circularly symmetric complex Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. The indicator function $\mathbb{I}_{\{\mathcal{A}\}}$ equals one when condition $\mathcal{A}$ is satisfied and zero otherwise. The notation $[x]_+ = \max(0, x)$ denotes the positive-part operator, while $\overset{c}{=}$ denotes equality up to an additive constant independent of the optimization variable.

D. Organization of the Paper

The remainder of the paper is organized as follows. Section II presents the system model, including the channel model, sparse activity representation, and received pilot transmission framework. Section III introduces the event-driven spatially correlated activity model and the corresponding spatial dependence structure among MTDs. Section IV develops the proposed CGAP-SBL framework, including conventional SBL, the Gaussian-inspired spatial interaction model, adaptive penalization, and the closed-form hyperparameter update. Section V establishes the theoretical properties of CGAP-SBL. Section VI presents the ANC interaction matrix construction and its integration into CGAP-SBL. Section VII analyzes the computational complexity of conventional SBL, CGAP-SBL, and CGAP-SBL-ANC. Section VIII presents the numerical results and performance analysis. Finally, Section IX concludes the paper.

II. System Model

This paper considers a single-cell MTC network consisting of a multi-antenna BS equipped with M antennas and N single-antenna MTDs uniformly distributed over a circular cell of radius R.

A. Channel Model

A block Rayleigh fading channel is assumed, where the channel coefficients remain constant during each transmission block. The channel vector between the i-th MTD and the BS is modeled as a zero-mean circularly symmetric complex Gaussian random vector. Specifically, the channel vector $\mathbf{h}_i$ for the i-th MTD is given by

$$\mathbf{h}_i \sim \mathcal{CN}(\mathbf{0}, g_i \mathbf{I}_M), \qquad \mathbf{h}_i \in \mathbb{C}^M, \tag{1}$$

where g_i represents the large-scale fading power coefficient accounting for path loss and shadowing associated with the i-th MTD. For simplicity, $g_i = 1$ is assumed, corresponding to normalized large-scale fading, which can be approximated when a power allocation algorithm is employed to compensate for propagation loss, as discussed in [37]. Under this assumption, the components of $\mathbf{h}_i$ are i.i.d. complex Gaussian random variables with zero mean and unit variance. The channel matrix for all N MTDs is defined as

$$\mathbf{H} = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \dots & \mathbf{h}_N \end{bmatrix} \in \mathbb{C}^{M\times N}. \tag{2}$$

Each element of $\mathbf{H}$ represents the fading coefficient between the m-th antenna of the BS and the i-th MTD, modeled as an i.i.d. complex Gaussian random variable

$$H_{m,i} \sim \mathcal{CN}(0, 1), \tag{3}$$

where m and i denote the antenna and MTD indices, respectively.

B. Sparse Activity Representation

Only a small subset of the N MTDs is assumed to be active during a given transmission block, resulting in a sparse transmission pattern. The activity pattern is generated according to the event-driven spatially correlated model introduced in Section III.

The true activation status of all MTDs is represented by the binary activity vector

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & \dots & a_N \end{bmatrix}^T \in \{0,1\}^N, \tag{4}$$

where a_i denotes the true activity state of the i-th MTD, with $a_i = 1$ indicating an active MTD and $a_i = 0$ indicating an inactive MTD. The realized number of active MTDs is given by $S = \sum_{i=1}^{N} a_i$.

The binary activity vector $\mathbf{a}$ is combined with the channel matrix $\mathbf{H}$ to form the effective channel matrix

$$\mathbf{Z} = \mathrm{diag}(\mathbf{a})\mathbf{H}^T \in \mathbb{C}^{N\times M}. \tag{5}$$

Thus, the i-th row of $\mathbf{Z}$ is $\mathbf{z}_i^T$, where $\mathbf{z}_i \in \mathbb{C}^M$ denotes the effective channel vector associated with the i-th MTD. When $a_i = 0$, $\mathbf{z}_i = \mathbf{0}$, whereas an active MTD with $a_i = 1$ contributes its corresponding channel vector. Consequently, $\mathbf{Z}$ exhibits row sparsity, with the nonzero rows corresponding to active MTDs.

C. Received Pilot Sequences

Non-orthogonal pilot sequences are assigned to the MTDs instead of orthogonal pilot sequences to reduce pilot resource consumption. Assigning mutually orthogonal pilot sequences to all N MTDs requires a pilot length of at least N, which may be impractical for systems with massive MTD connectivity. In contrast, non-orthogonal pilot sequences enable pilot transmission with $L < N$ by exploiting the sparse activity pattern. Each MTD is assigned a known pilot sequence of length L, consisting of quadrature phase-shift keying (QPSK) symbols. The pilot matrix $\mathbf{\Theta} \in \mathbb{C}^{L\times N}$, which serves as the sensing matrix for CS-based recovery, contains the pilot sequences assigned to all MTDs and is known to the BS.

The received signal during the pilot transmission phase, $\mathbf{Y} \in \mathbb{C}^{L\times M}$, is given by

$$\mathbf{Y} = \mathbf{\Theta}\mathbf{Z} + \mathbf{W}, \tag{6}$$

where $\mathbf{W} \in \mathbb{C}^{L\times M}$ represents additive white Gaussian noise (AWGN) with i.i.d. elements modeled as

$$W_{l,m} \sim \mathcal{CN}(0,\sigma^2), \quad l = 1,\dots,L, \quad m = 1,\dots,M, \tag{7}$$

where σ^2 denotes the noise variance.

III. Event-Driven Spatially Correlated Activity Model

A. Event-Based MTD Activation

The sparse MTD activity introduced in Section II is modeled through spatially distributed triggering events. It is assumed that V random events occur within the coverage area, with their locations uniformly distributed within a disk of radius R centered at the BS. Each event influences the activation probability of nearby MTDs according to their spatial proximity.

As illustrated in Fig. 1, MTDs located closer to an event epicenter have a higher probability of activation. Consequently, nearby MTDs may exhibit correlated activity because they can be influenced by the same event.

The probability that the v-th event activates the i-th MTD is denoted by P_{iv} and is determined by the distance d_{iv} between the MTD and the corresponding event epicenter. Accordingly, the distance-dependent activation probability is defined through the function

$$f_v : [0,\infty) \to [0,1], \quad d_{iv} \mapsto f_v(d_{iv}) = P_{iv}. \tag{8}$$

As d_{iv} increases, $f(d_{iv})$ decreases, reflecting a lower probability of activation for the i-th MTD. Conversely, if an event occurs near the i-th MTD, its activation probability approaches one [8]. This behavior is modeled using an exponential decay function

$$P_{iv} = \frac{\exp\left(-\frac{d_{iv}}{\kappa_v}\right) - \exp\left(-\frac{D_v}{\kappa_v}\right)}{1 - \exp\left(-\frac{D_v}{\kappa_v}\right)} \cdot \mathbb{I}_{\{d_{iv}\le D_v\}}, \tag{9}$$

where $\kappa_v > 0$ denotes the spatial decay parameter associated with the v-th event, while $D_v > 0$ denotes its maximum influence distance. For simplicity, all events are assumed to share common parameters, i.e., $\kappa_v = \kappa$ and $D_v = D$ for all v. Under this assumption, all event-dependent activation functions coincide, i.e., $f_v = f$ for all v.

The exponential decay function captures the rapid decrease in activation probability with distance, while the maximum influence distance ensures that events do not activate MTDs located outside their influence regions.

Note that MTDs within the coverage disk may be impacted by several events. To determine the activation probability of each MTD, the effects of all events must therefore be considered. Assuming that the triggering effects of different events on a given MTD are conditionally independent, given the event locations, the probability of the i-th MTD being activated by at least one of the V events can be expressed as

$$\begin{aligned} P_i &= \Pr\{i\text{-th MTD is activated by at least one event}\} \\ &= 1 - \Pr\{i\text{-th MTD is not activated by any event}\} \\ &= 1 - \prod_{v=1}^{V} (1 - P_{iv}) \\ &= 1 - \prod_{v=1}^{V} \left[1 - \frac{\exp\left(-\frac{d_{iv}}{\kappa}\right) - \exp\left(-\frac{D}{\kappa}\right)}{1 - \exp\left(-\frac{D}{\kappa}\right)} \cdot \mathbb{I}_{\{d_{iv}\le D\}}\right]. \end{aligned} \tag{10}$$

Here, P_i is conditioned on the realized event locations. The activity state of each MTD is subsequently generated according to its corresponding activation probability. Since nearby MTDs can be influenced by the same randomly located events, the event-driven activation mechanism can induce spatial dependence among their activity states across traffic realizations.

This model provides a practical representation of event-driven MTD activity, with activation probabilities decreasing smoothly with distance within each event's influence region while preserving the stochastic nature of MTD activation.

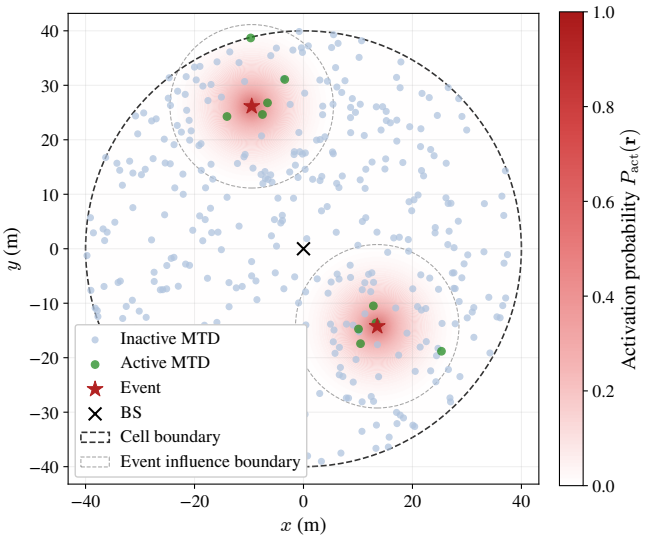


Fig. 1. Illustration of the event-driven spatially correlated MTD activity model. The background shows the spatial activation probability $P_{\text{act}}(\mathbf{r}) = 1 - \prod_{v=1}^{V}[1 - f(d_v(\mathbf{r}))]$, where $\mathbf{r} = [x, y]^T$ denotes a spatial location in Cartesian coordinates relative to the BS (in meters), $d_v(\mathbf{r})$ is its distance to the v-th event, and $f(\cdot)$ is defined in (9). At the location $\mathbf{r}_i$ of MTD i, $P_{\text{act}}(\mathbf{r}_i) = P_i$.

B. Spatial Correlation Model

The event-driven activation mechanism described above can induce spatial dependence among nearby MTDs. To represent this dependence, a distance-based spatial correlation model is adopted, in which MTDs located closer to each other are assigned higher spatial dependence coefficients. These coefficients decrease with increasing separation distance and become zero beyond a maximum correlation distance U.

The modeled spatial dependence matrix is denoted by $\mathbf{C} \in \mathbb{R}^{N\times N}$, where the (i, j)-th element C_{ij} represents a distance-based measure of spatial association between the i-th and j-th MTDs. It is defined as

$$C_{ij} = \frac{\exp\left(-\frac{u_{ij}}{\rho}\right) - \exp\left(-\frac{U}{\rho}\right)}{1 - \exp\left(-\frac{U}{\rho}\right)} \cdot \mathbb{I}_{\{u_{ij}\le U\}}, \tag{11}$$

where u_{ij} denotes the Euclidean distance between the i-th and j-th MTDs, $\rho > 0$ is the spatial correlation decay parameter, and $U > 0$ represents the maximum distance within which two MTDs are assigned nonzero spatial dependence.

The modeled coefficients satisfy $0 \le C_{ij} \le 1$, with $C_{ii} = 1$ and $C_{ij} = C_{ji}$. The value of C_{ij} decreases as the separation distance u_{ij} increases, while MTDs separated by more than U are assigned zero spatial dependence. It should be noted that $\mathbf{C}$ is a distance-based approximation of spatial activity relationships rather than an empirically estimated statistical correlation matrix of the binary activity states. The resulting matrix provides the spatial information used to construct the interaction matrix within the proposed CGAP-SBL-ANC algorithm.

IV. Correlation-guided adaptive penalized Sparse Bayesian Learning

A. Conventional Sparse Bayesian Learning

SBL provides a probabilistic framework for recovering sparse signals by introducing a set of continuous hyperparameters that control the prior variances of the unknown signal coefficients [23]. In the MMV setting, SBL exploits the common support shared across measurement vectors to estimate the row-sparse effective channel matrix $\mathbf{Z}$ from the received pilot observations $\mathbf{Y}$ [24].

Following the received signal model in (6), the observation at the m-th BS antenna is expressed as

$$\mathbf{y}^{(m)} = \mathbf{\Theta}\mathbf{z}^{(m)} + \mathbf{w}^{(m)}, \tag{12}$$

where $\mathbf{y}^{(m)} \in \mathbb{C}^L$, $\mathbf{z}^{(m)} \in \mathbb{C}^N$, and $\mathbf{w}^{(m)} \in \mathbb{C}^L$ denote the m-th columns of $\mathbf{Y}$, $\mathbf{Z}$, and $\mathbf{W}$, respectively. Given $\mathbf{z}^{(m)}$, the observation follows the complex Gaussian distribution

$$\mathbf{y}^{(m)} \mid \mathbf{z}^{(m)} \sim \mathcal{CN}\left(\mathbf{\Theta}\mathbf{z}^{(m)}, \sigma^2\mathbf{I}_L\right). \tag{13}$$

To promote row sparsity, conventional SBL assigns an independent zero-mean complex Gaussian prior to each effective channel vector $\mathbf{z}_i$, such that

$$\mathbf{z}_i \mid \gamma_i \sim \mathcal{CN}(\mathbf{0}, \gamma_i\mathbf{I}_M), \tag{14}$$

where $\gamma_i \ge 0$ is the variance hyperparameter associated with the i-th MTD. The hyperparameter vector is denoted by $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_N]^T \in \mathbb{R}_+^N$, and its corresponding diagonal matrix is defined as $\mathbf{\Gamma} = \text{diag}(\boldsymbol{\gamma})$. Small values of γ_i concentrate the prior distribution around zero, thereby promoting the suppression of inactive MTDs, whereas larger values allow nonzero effective channel coefficients. A zero variance corresponds to a degenerate Gaussian distribution concentrated at zero. The subsequent density-based EM derivation assumes strictly positive hyperparameters, with the zero-variance case understood as a limiting case.

By marginalizing the effective channel coefficients, the received observation at each BS antenna follows

$$\mathbf{y}^{(m)} \mid \boldsymbol{\gamma} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Pi}), \tag{15}$$

where the observation covariance matrix is

$$\mathbf{\Pi} = \mathbf{\Theta}\mathbf{\Gamma}\mathbf{\Theta}^H + \sigma^2\mathbf{I}_L. \tag{16}$$

Since the observations are conditionally independent across the M BS antennas, the marginal log-likelihood is given by

$$\log p(\mathbf{Y} \mid \boldsymbol{\gamma}) \overset{c}{=} -M \log\det(\mathbf{\Pi}) - \text{Tr}\left(\mathbf{Y}^H\mathbf{\Pi}^{-1}\mathbf{Y}\right). \tag{17}$$

Conventional SBL estimates the hyperparameters by maximizing this marginal likelihood, commonly referred to as type-II maximum likelihood estimation.

The EM algorithm provides an iterative approach to hyperparameter estimation by treating $\mathbf{Z}$ as a latent variable. At iteration k, given the previous hyperparameter estimate $\boldsymbol{\gamma}^{[k-1]}$, the E-step evaluates the posterior distribution $p(\mathbf{Z} \mid \mathbf{Y}, \boldsymbol{\gamma}^{[k-1]})$. The posterior mean matrix $\boldsymbol{\mu}^{[k]}$ and covariance matrix $\mathbf{\Sigma}^{[k]}$ are obtained as

$$\boldsymbol{\mu}^{[k]} = \mathbf{\Gamma}^{[k-1]}\mathbf{\Theta}^H\left(\mathbf{\Pi}^{[k-1]}\right)^{-1}\mathbf{Y}, \tag{18}$$

$$\boldsymbol{\Sigma}^{[k]} = \boldsymbol{\Gamma}^{[k-1]} - \boldsymbol{\Gamma}^{[k-1]}\boldsymbol{\Theta}^H \left(\boldsymbol{\Pi}^{[k-1]}\right)^{-1} \boldsymbol{\Theta}\boldsymbol{\Gamma}^{[k-1]}, \quad (19)$$

where $\boldsymbol{\Gamma}^{[k-1]} = \mathrm{diag}(\boldsymbol{\gamma}^{[k-1]})$ and $\boldsymbol{\Pi}^{[k-1]} = \boldsymbol{\Theta}\boldsymbol{\Gamma}^{[k-1]}\boldsymbol{\Theta}^H + \sigma^2\mathbf{I}_L$. The vector $\boldsymbol{\mu}_i^{[k]} \in \mathbb{C}^M$ denotes the transpose of the i-th row of $\boldsymbol{\mu}^{[k]}$, while $\Sigma_{ii}^{[k]}$ denotes the i-th diagonal element of $\boldsymbol{\Sigma}^{[k]}$.

In the M-step, the hyperparameters are updated by maximizing the EM auxiliary function

$$Q(\boldsymbol{\gamma}; \boldsymbol{\gamma}^{[k-1]}) = \mathbb{E}_{\mathbf{Z}|\mathbf{Y},\boldsymbol{\gamma}^{[k-1]}}[\log p(\mathbf{Y}, \mathbf{Z} \mid \boldsymbol{\gamma})]. \quad (20)$$

Since the conditional likelihood $p(\mathbf{Y} \mid \mathbf{Z})$ does not depend on the candidate hyperparameters $\boldsymbol{\gamma}$, the terms relevant to the M-step arise from the prior distribution $p(\mathbf{Z} \mid \boldsymbol{\gamma})$. Consequently, the auxiliary function can be expressed as

$$Q(\boldsymbol{\gamma}; \boldsymbol{\gamma}^{[k-1]}) \overset{c}{=} -M \sum_{i=1}^{N} \left(\log \gamma_i + \frac{\eta_i^{[k]}}{\gamma_i} \right), \quad (21)$$

where

$$\eta_i^{[k]} = \Sigma_{ii}^{[k]} + \frac{1}{M} \left\| \boldsymbol{\mu}_i^{[k]} \right\|_2^2 \quad (22)$$

denotes the normalized posterior second-order statistic associated with the i-th MTD.

Maximizing (21) with respect to each $\gamma_i > 0$ yields

$$\frac{\partial Q}{\partial \gamma_i} = -\frac{M}{\gamma_i} + \frac{M\eta_i^{[k]}}{\gamma_i^2} = 0, \quad (23)$$

which gives the conventional SBL hyperparameter update

$$\boxed{\gamma_i^{[k]} = \eta_i^{[k]} = \Sigma_{ii}^{[k]} + \frac{1}{M} \left\| \boldsymbol{\mu}_i^{[k]} \right\|_2^2}. \quad (24)$$

The updated hyperparameters determine the prior variances used in the subsequent E-step. Through repeated posterior estimation and hyperparameter updates, conventional SBL promotes row sparsity in the effective channel matrix. However, the independent prior structure in (14) does not explicitly account for spatial dependence among MTD activity states. This motivates the incorporation of spatial interaction information into the hyperparameter estimation process, as developed in the following subsections.

B. Gaussian-Inspired Spatial Interaction Model

In conventional SBL, the variance hyperparameters $\boldsymbol{\gamma}$ are estimated independently in the M-step, as shown in (21). Although this formulation promotes row sparsity in the effective channel matrix, it does not explicitly incorporate spatial dependence among MTD activity states. In event-driven MTC networks, nearby MTDs may be influenced by common triggering events, resulting in spatially correlated activation patterns. Such relationships can provide additional information for estimating the hyperparameters associated with individual MTDs.

To incorporate this information, a Gaussian-inspired quadratic interaction model is considered over the continuous, non-negative hyperparameter vector $\boldsymbol{\gamma}$. The spatial interaction energy is defined as

$$\mathcal{R}_{\boldsymbol{\Omega}}(\boldsymbol{\gamma}) = \frac{1}{2}\boldsymbol{\gamma}^T\boldsymbol{\Omega}\boldsymbol{\gamma}, \quad (25)$$

where $\boldsymbol{\Omega} \in \mathbb{R}^{N\times N}$ is a symmetric spatial interaction matrix whose elements characterize the modeled relationships among the MTD hyperparameters. Expanding (25) yields

$$\mathcal{R}_{\boldsymbol{\Omega}}(\boldsymbol{\gamma}) = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\Omega_{ij}\gamma_i\gamma_j. \quad (26)$$

The quadratic structure is motivated by the interaction terms appearing in multivariate Gaussian distributions. However, $\boldsymbol{\Omega}$ is not generally required to be positive definite, and therefore (25) is treated as a Gaussian-inspired interaction energy rather than the negative log-density of a necessarily proper Gaussian hyperprior.

Since $\boldsymbol{\Omega}$ is symmetric, the gradient of the interaction energy with respect to $\boldsymbol{\gamma}$ is

$$\nabla_{\boldsymbol{\gamma}}\mathcal{R}_{\boldsymbol{\Omega}}(\boldsymbol{\gamma}) = \boldsymbol{\Omega}\boldsymbol{\gamma}, \quad (27)$$

whose i-th component is given by

$$\frac{\partial \mathcal{R}_{\boldsymbol{\Omega}}(\boldsymbol{\gamma})}{\partial \gamma_i} = \sum_{j=1}^{N}\Omega_{ij}\gamma_j. \quad (28)$$

Thus, the interaction associated with each MTD depends on the hyperparameter estimates of other MTDs through the corresponding entries of $\boldsymbol{\Omega}$. Unlike conventional SBL, where the M-step separates across the candidate hyperparameters, this interaction introduces coupling among them.

Directly incorporating the quadratic interaction into the SBL objective would generally produce a coupled hyperparameter optimization problem. To retain computationally tractable updates, the proposed CGAP-SBL framework evaluates the interaction gradient using the hyperparameter estimates from the preceding iteration and employs its non-negative components to construct adaptive penalization coefficients. The resulting penalized EM formulation and its closed-form hyperparameter update are developed in the following subsections.

C. Adaptive Penalized EM Formulation

The Gaussian-inspired spatial interaction model introduced in the preceding subsection provides a mechanism for incorporating spatial dependence among MTDs into the hyperparameter estimation process. However, directly including the quadratic interaction energy in the conventional SBL objective introduces coupling among the hyperparameters, making the resulting optimization problem more challenging to solve. To preserve the computational tractability of the conventional EM-based updates, the proposed CGAP-SBL framework employs an iteration-dependent spatial penalization strategy.

At iteration k, the spatial interaction gradient in (27) is evaluated using the hyperparameter estimates from the preceding iteration, $\boldsymbol{\gamma}^{[k-1]}$. The adaptive penalty coefficient associated with the i-th MTD is defined as

$$\lambda_i^{[k]} = \left[(\boldsymbol{\Omega}\boldsymbol{\gamma}^{[k-1]})_i \right]_+ = \max\left(0, \sum_{j=1}^{N}\Omega_{ij}\gamma_j^{[k-1]}\right). \quad (29)$$

Accordingly, $\lambda_i^{[k]} \geq 0$ for all MTDs. The penalty coefficients are determined by the spatial interaction matrix $\boldsymbol{\Omega}$ and the

preceding hyperparameter estimates, allowing the spatial penalization to adapt across iterations.

The non-negative restriction ensures that the spatial interaction contributes only an additional penalty to the conventional SBL objective. When the evaluated interaction gradient is non-positive, the corresponding penalty is set to zero, and the conventional SBL hyperparameter update is recovered for that MTD. Furthermore, non-negative penalty coefficients prevent the linear penalization term from making the resulting maximization problem unbounded above as a hyperparameter increases.

Using the EM auxiliary function in (21), the iteration-specific penalized objective is defined as

$$\mathcal{J}_k(\boldsymbol{\gamma}) = Q(\boldsymbol{\gamma}; \boldsymbol{\gamma}^{[k-1]}) - \sum_{i=1}^{N} \lambda_i^{[k]} \gamma_i. \tag{30}$$

Substituting (21) into (30) gives

$$\mathcal{J}_k(\boldsymbol{\gamma}) \overset{c}{=} -M \sum_{i=1}^{N} \left(\log \gamma_i + \frac{\eta_i^{[k]}}{\gamma_i} + \frac{\lambda_i^{[k]}}{M} \gamma_i \right). \tag{31}$$

Here, the posterior second-order statistics $\eta_i^{[k]}$ are obtained from the E-step, while the penalty coefficients $\lambda_i^{[k]}$ are computed using $\boldsymbol{\gamma}^{[k-1]}$. Both quantities remain fixed during the current M-step.

Consequently, the hyperparameter estimation problem at iteration k is formulated as

$$\boldsymbol{\gamma}^{[k]} = \underset{\boldsymbol{\gamma} \in \mathbb{R}_{++}^{N}}{\arg\max}\ \mathcal{J}_k(\boldsymbol{\gamma}). \tag{32}$$

Since the adaptive penalty coefficients are fixed during each M-step, the objective in (31) is separable with respect to the candidate hyperparameters. Therefore, each $\gamma_i^{[k]}$ can be updated independently through a scalar optimization problem.

Unlike conventional SBL, which estimates the hyperparameters without explicitly incorporating spatial activity relationships, the proposed formulation introduces spatially adaptive penalization through $\boldsymbol{\Omega}$. The interaction matrix is not restricted to a particular construction, allowing CGAP-SBL to accommodate different spatial dependence models. Moreover, the formulation optimizes an iteration-specific penalized EM objective rather than assuming that the Gaussian-inspired interaction defines a proper hyperparameter prior. The corresponding closed-form hyperparameter update is derived in the following subsection.

D. Closed-Form Hyperparameter Update

The adaptive penalized EM formulation in (31) enables independent optimization of the variance hyperparameters during each M-step. Since the penalty coefficients $\lambda_i^{[k]}$ and posterior second-order statistics $\eta_i^{[k]}$ are fixed at iteration k, the optimization problem in (32) can be decomposed into N scalar subproblems. For the i-th MTD, the corresponding objective is

$$j_i^{[k]}(\gamma_i) = -M \log \gamma_i - \frac{M \eta_i^{[k]}}{\gamma_i} - \lambda_i^{[k]} \gamma_i, \tag{33}$$

where $\gamma_i > 0$, $\eta_i^{[k]} > 0$, and $\lambda_i^{[k]} \geq 0$.

Differentiating (33) with respect to γ_i gives

$$\frac{\partial j_i^{[k]}}{\partial \gamma_i} = -\frac{M}{\gamma_i} + \frac{M \eta_i^{[k]}}{\gamma_i^2} - \lambda_i^{[k]}. \tag{34}$$

Setting the derivative to zero and multiplying by γ_i^2 / M yields

$$\frac{\lambda_i^{[k]}}{M} \gamma_i^2 + \gamma_i - \eta_i^{[k]} = 0. \tag{35}$$

For $\lambda_i^{[k]} > 0$, the positive root of (35) is

$$\gamma_i^{[k]} = \frac{M}{2\lambda_i^{[k]}} \left(\sqrt{1 + \frac{4\lambda_i^{[k]} \eta_i^{[k]}}{M}} - 1 \right). \tag{36}$$

Although this expression is mathematically valid, direct evaluation may suffer from subtractive cancellation when the penalty coefficient is small. Rationalizing the numerator gives the numerically stable expression

$$\boxed{\gamma_i^{[k]} = \frac{2\eta_i^{[k]}}{1 + \sqrt{1 + \frac{4\lambda_i^{[k]} \eta_i^{[k]}}{M}}}.} \tag{37}$$

Unlike (36), (37) remains well-defined when $\lambda_i^{[k]} = 0$. In this case, the update reduces exactly to the conventional SBL update in (24), i.e.,

$$\gamma_i^{[k]} = \eta_i^{[k]}, \qquad \lambda_i^{[k]} = 0. \tag{38}$$

Consequently, the proposed CGAP-SBL update can be applied using the non-negative adaptive penalty coefficients in (29) without requiring a separate numerical approximation for non-positive spatial interactions. When the evaluated spatial interaction is positive, the update introduces additional shrinkage relative to conventional SBL. Otherwise, the corresponding penalty vanishes and the conventional SBL update is recovered.

The update in (37) is the unique positive maximizer of the scalar penalized EM subproblem under the stated assumptions, as formally established in Section V. Importantly, this result concerns the iteration-specific M-step and does not, by itself, establish global convergence of the complete CGAP-SBL algorithm.

E. Joint Activity Detection and Channel Estimation

The proposed CGAP-SBL framework performs JADCE by iteratively estimating the effective channel matrix and updating the variance hyperparameters using the received pilot observations and spatial interaction information. Given the pilot matrix $\boldsymbol{\Theta}$, received signal matrix $\mathbf{Y}$, noise variance σ^2, and a symmetric spatial interaction matrix $\boldsymbol{\Omega}$, the algorithm is initialized with strictly positive hyperparameters $\boldsymbol{\gamma}^{[0]} \in \mathbb{R}_{++}^{N}$.

At each iteration k, the posterior mean $\boldsymbol{\mu}^{[k]}$ and covariance $\boldsymbol{\Sigma}^{[k]}$ are obtained using (18) and (19), respectively. The posterior second-order statistics $\eta_i^{[k]}$ are then evaluated according to (22). Subsequently, the adaptive penalty coefficients $\lambda_i^{[k]}$ are computed using (29), and the hyperparameters are updated through the closed-form expression in (37). This procedure is

repeated until a prescribed convergence tolerance is satisfied or the maximum number of iterations is reached.

The relative change in the hyperparameter vector is used as a stopping criterion, given by

$$\frac{\left\|\boldsymbol{\gamma}^{[k]}-\boldsymbol{\gamma}^{[k-1]}\right\|_2}{\left\|\boldsymbol{\gamma}^{[k-1]}\right\|_2}<\delta, \tag{39}$$

where $\delta>0$ denotes the convergence tolerance. The maximum number of iterations is denoted by K.

Following termination, the final hyperparameter estimates are used to obtain the posterior mean of the effective channel matrix, which serves as the channel estimate

$$\widehat{\mathbf{Z}}=\widehat{\boldsymbol{\Gamma}}\boldsymbol{\Theta}^H\widehat{\boldsymbol{\Pi}}^{-1}\mathbf{Y}, \tag{40}$$

where $\widehat{\boldsymbol{\Gamma}}=\mathrm{diag}(\widehat{\boldsymbol{\gamma}})$, $\widehat{\boldsymbol{\gamma}}$ denotes the final hyperparameter vector, and $\widehat{\boldsymbol{\Pi}}=\boldsymbol{\Theta}\widehat{\boldsymbol{\Gamma}}\boldsymbol{\Theta}^H+\sigma^2\mathbf{I}_L$.

The activity status of the i-th MTD is determined by comparing its final variance hyperparameter against a detection threshold $\tau>0$, such that

$$\widehat{a}_i=\mathbb{I}_{\{\widehat{\gamma}_i\geq\tau\}},\qquad i=1,\ldots,N, \tag{41}$$

where $\widehat{a}_i=1$ indicates that the i-th MTD is detected as active, while $\widehat{a}_i=0$ indicates that it is detected as inactive. The estimated activity vector is therefore $\widehat{\mathbf{a}}=[\widehat{a}_1,\ldots,\widehat{a}_N]^T$.

The threshold τ controls the trade-off between missed detections and false alarms and is selected according to the detection requirements. Importantly, the continuous hyperparameters are used directly for posterior channel estimation, while the activity threshold is applied separately to obtain the binary activity decisions.

The complete CGAP-SBL procedure is summarized in Algorithm 1. The framework accepts a general symmetric interaction matrix $\boldsymbol{\Omega}$, allowing spatial dependence information to be incorporated through different matrix constructions. The ANC-based construction employed in this work is developed in Section VI.

Algorithm 1 CGAP-SBL

Require: $\mathbf{Y}$, $\boldsymbol{\Theta}$, σ^2, symmetric $\boldsymbol{\Omega}$, $\boldsymbol{\gamma}^{[0]}>0$, τ, δ, and K
Ensure: Estimated activity vector $\widehat{\mathbf{a}}$ and effective channel matrix $\widehat{\mathbf{Z}}$

for $k=1,\ldots,K$ **do**
  Compute $\boldsymbol{\Gamma}^{[k-1]}=\mathrm{diag}(\boldsymbol{\gamma}^{[k-1]})$.
  Compute $\boldsymbol{\Pi}^{[k-1]}=\boldsymbol{\Theta}\boldsymbol{\Gamma}^{[k-1]}\boldsymbol{\Theta}^H+\sigma^2\mathbf{I}_L$.
  Compute $\boldsymbol{\mu}^{[k]}$ and $\boldsymbol{\Sigma}^{[k]}$ using (18) and (19).
  for $i=1,\ldots,N$ **do**
    Compute $\eta_i^{[k]}$ using (22).
  end for
  Compute $\boldsymbol{\lambda}^{[k]}=[\boldsymbol{\Omega}\boldsymbol{\gamma}^{[k-1]}]_+$ elementwise.
  for $i=1,\ldots,N$ **do**
    Update $\gamma_i^{[k]}$ using (37).
  end for
  if $\|\boldsymbol{\gamma}^{[k]}-\boldsymbol{\gamma}^{[k-1]}\|_2/\|\boldsymbol{\gamma}^{[k-1]}\|_2<\delta$ **then**
    break
  end if
end for
Set $\widehat{\boldsymbol{\gamma}}=\boldsymbol{\gamma}^{[k]}$.
Compute $\widehat{\mathbf{Z}}$ using (40).
for $i=1,\ldots,N$ **do**
  Set $\widehat{a}_i=\mathbb{I}_{\{\widehat{\gamma}_i\geq\tau\}}$.
end for
return $\widehat{\mathbf{a}}$, $\widehat{\mathbf{Z}}$

V. Theoretical Properties of CGAP-SBL

A. Existence and Uniqueness of the Hyperparameter Update

The proposed CGAP-SBL framework updates the variance hyperparameters by maximizing the iteration-specific penalized EM objective in (31). Since the objective is separable with respect to the candidate hyperparameters, the existence and uniqueness of its maximizer can be established by analyzing the corresponding scalar subproblems.

Proposition 1: For a fixed posterior second-order statistic $\eta_i^{[k]}>0$, adaptive penalty coefficient $\lambda_i^{[k]}\geq 0$, and number of BS antennas $M\geq 1$, the scalar penalized EM objective in (33) admits a unique global maximizer over $\gamma_i>0$, given by

$$\gamma_i^{[k]}=\frac{2\eta_i^{[k]}}{1+\sqrt{1+\frac{4\lambda_i^{[k]}\eta_i^{[k]}}{M}}}. \tag{42}$$

Proof: For notational simplicity, the iteration and device indices are omitted throughout the proof. Consider the scalar objective

$$j(\gamma)=-M\log\gamma-\frac{M\eta}{\gamma}-\lambda\gamma, \tag{43}$$

where $\gamma>0$, $\eta>0$, and $\lambda\geq 0$. Its first derivative is

$$j'(\gamma)=\frac{M}{\gamma^2}\left(\eta-\gamma-\frac{\lambda}{M}\gamma^2\right). \tag{44}$$

Define the function

$$h(\gamma)=\gamma+\frac{\lambda}{M}\gamma^2. \tag{45}$$

Since

$$h'(\gamma)=1+\frac{2\lambda}{M}\gamma>0, \tag{46}$$

$h(\gamma)$ is strictly increasing for all $\gamma>0$. Furthermore, $h(0)=0<\eta$ and $\lim_{\gamma\to\infty}h(\gamma)=\infty$. Therefore, the stationary condition $h(\gamma)=\eta$ admits exactly one positive solution, denoted by $\gamma^\star$.

From (44), it follows that $j'(\gamma)>0$ for $0<\gamma<\gamma^\star$ and $j'(\gamma)<0$ for $\gamma>\gamma^\star$. Consequently, $j(\gamma)$ is strictly increasing before $\gamma^\star$ and strictly decreasing afterward. Thus, $\gamma^\star$ is the unique global maximizer of the scalar objective over $\gamma>0$.

Solving the stationary condition gives

$$\frac{\lambda}{M}(\gamma^\star)^2+\gamma^\star-\eta=0. \tag{47}$$

For $\lambda>0$, the positive root can be rationalized to obtain

$$\gamma^\star=\frac{2\eta}{1+\sqrt{1+\frac{4\lambda\eta}{M}}}. \tag{48}$$

When $\lambda=0$, the stationary condition reduces to $\gamma^\star=\eta$, which is also obtained directly from (48). Therefore, the proposed closed-form expression provides the unique positive global maximizer for all $\lambda\geq 0$. ∎

Proposition 1 establishes that each iteration-specific penalized EM subproblem admits a unique positive global maximizer under the stated assumptions. Since the objective in (31) is separable, independently applying the closed-form update to all N hyperparameters yields the unique global maximizer of the corresponding iteration-specific penalized M-step. This result holds for any symmetric spatial interaction matrix $\mathbf{\Omega}$ because the non-negative penalty coefficients are fixed during the optimization. However, it does not imply global optimality or convergence of the complete iterative CGAP-SBL procedure, in which the posterior statistics and penalty coefficients are recomputed at successive iterations.

B. Adaptive Shrinkage and Sensitivity Properties

The adaptive penalty coefficients introduced in (29) modify the conventional SBL hyperparameter updates according to the spatial interaction information. The resulting shrinkage behavior can be characterized analytically using the closed-form update in (37).

Proposition 2: For fixed $\eta_i^{[k]} > 0$, $\lambda_i^{[k]} \geq 0$, and $M \geq 1$, the CGAP-SBL hyperparameter update satisfies

$$0 < \gamma_i^{[k]} \leq \eta_i^{[k]}, \tag{49}$$

where equality holds if and only if $\lambda_i^{[k]} = 0$. Furthermore, for fixed $\eta_i^{[k]}$, the updated hyperparameter is a strictly decreasing function of $\lambda_i^{[k]}$, with

$$\frac{\partial \gamma_i^{[k]}}{\partial \lambda_i^{[k]}} = -\frac{(\gamma_i^{[k]})^2}{M + 2\lambda_i^{[k]}\gamma_i^{[k]}} < 0. \tag{50}$$

Proof: For notational simplicity, the iteration and device indices are omitted. From the stationary condition in (35), the optimal hyperparameter satisfies

$$\eta = \gamma + \frac{\lambda}{M}\gamma^2. \tag{51}$$

Since $\gamma > 0$, $\lambda \geq 0$, and $M > 0$, it follows that

$$\eta - \gamma = \frac{\lambda}{M}\gamma^2 \geq 0. \tag{52}$$

Therefore, $0 < \gamma \leq \eta$, with equality if and only if $\lambda = 0$.

To establish the sensitivity to the penalty coefficient, implicitly differentiate (51) with respect to λ, while holding η fixed:

$$0 = \frac{\partial \gamma}{\partial \lambda} + \frac{\gamma^2}{M} + \frac{2\lambda\gamma}{M}\frac{\partial \gamma}{\partial \lambda}. \tag{53}$$

Rearranging gives

$$\frac{\partial \gamma}{\partial \lambda} = -\frac{\gamma^2}{M + 2\lambda\gamma}. \tag{54}$$

Since the numerator is strictly positive and the denominator is positive for all $\gamma > 0$ and $\lambda \geq 0$, the derivative is strictly negative. Hence, for fixed posterior statistics, increasing the adaptive penalty coefficient strictly decreases the updated hyperparameter. ∎

Proposition 2 establishes that the proposed adaptive penalization introduces additional shrinkage relative to the conventional SBL update for fixed posterior statistics. When $\lambda_i^{[k]} = 0$, the update coincides with conventional SBL, whereas a positive penalty produces a smaller hyperparameter estimate. The magnitude of this shrinkage is controlled by the penalty coefficient derived from the spatial interaction information.

The sensitivity of the update to the posterior second-order statistic can also be characterized.

Proposition 3: For fixed $\lambda_i^{[k]} \geq 0$ and $M \geq 1$, the CGAP-SBL hyperparameter update is a strictly increasing function of $\eta_i^{[k]} > 0$ and satisfies

$$\frac{\partial \gamma_i^{[k]}}{\partial \eta_i^{[k]}} = \frac{M}{M + 2\lambda_i^{[k]}\gamma_i^{[k]}}, \tag{55}$$

with

$$0 < \frac{\partial \gamma_i^{[k]}}{\partial \eta_i^{[k]}} \leq 1. \tag{56}$$

Equality in the upper bound holds if and only if $\lambda_i^{[k]} = 0$.

Proof: Again, consider the stationary condition in (51). Implicit differentiation with respect to η, while holding λ fixed, yields

$$1 = \frac{\partial \gamma}{\partial \eta} + \frac{2\lambda\gamma}{M}\frac{\partial \gamma}{\partial \eta}. \tag{57}$$

Therefore,

$$\frac{\partial \gamma}{\partial \eta} = \frac{M}{M + 2\lambda\gamma}. \tag{58}$$

Since $\lambda \geq 0$, $\gamma > 0$, and $M > 0$, the derivative is strictly positive and does not exceed one. The upper bound is attained precisely when $\lambda = 0$. ∎

Proposition 3 shows that the updated hyperparameters remain responsive to the posterior second-order statistics, even under spatial penalization. However, positive penalties reduce the sensitivity of the update relative to conventional SBL, for which $\partial \gamma_i^{[k]} / \partial \eta_i^{[k]} = 1$. Together, Propositions 2 and 3 establish that CGAP-SBL introduces a controllable shrinkage mechanism while preserving the monotonic relationship between posterior evidence and the updated variance hyperparameters.

C. Reduction to Conventional SBL

An important property of the proposed CGAP-SBL framework is its ability to recover conventional SBL when spatial penalization is disabled. This relationship establishes conventional SBL as a special case of the proposed framework and provides a reference for examining the effect of spatial interaction information.

Proposition 4: Consider the CGAP-SBL framework with strictly positive initial hyperparameters $\boldsymbol{\gamma}^{[0]} \in \mathbb{R}^N_{++}$. If $\mathbf{\Omega} = \mathbf{0}$, the adaptive penalty coefficients vanish at every iteration, and the resulting CGAP-SBL hyperparameter updates are identical to those of conventional SBL. Furthermore, for fixed $\eta_i^{[k]} > 0$, the CGAP-SBL update is continuous at $\lambda_i^{[k]} = 0$ and satisfies

$$\lim_{\lambda_i^{[k]} \to 0^+} \gamma_i^{[k]} = \eta_i^{[k]}. \tag{59}$$

Proof: From the adaptive penalty definition in (29), setting $\mathbf{\Omega} = \mathbf{0}$ gives

$$\lambda_i^{[k]} = \left[(\mathbf{\Omega}\boldsymbol{\gamma}^{[k-1]})_i \right]_+ = 0, \quad \forall i, k. \tag{60}$$

Substituting $\lambda_i^{[k]} = 0$ into the CGAP-SBL update in (37) yields

$$\gamma_i^{[k]} = \frac{2\eta_i^{[k]}}{1+\sqrt{1}} = \eta_i^{[k]}, \tag{61}$$

which is identical to the conventional SBL update in (24).

Moreover, the posterior mean and covariance computations in (18) and (19) are common to both algorithms. Therefore, given identical initialization, noise variance, and received observations, both algorithms produce identical posterior statistics at the first iteration. Since their hyperparameter updates also coincide when $\mathbf{\Omega} = \mathbf{0}$, the subsequent hyperparameter vectors remain identical. By induction, this equivalence holds at every iteration, and both algorithms yield the same estimates under identical stopping and decision rules.

Finally, for fixed $\eta_i^{[k]} > 0$, taking the limit of (37) as $\lambda_i^{[k]} \to 0^+$ gives

$$\lim_{\lambda_i^{[k]} \to 0^+} \frac{2\eta_i^{[k]}}{1+\sqrt{1+\frac{4\lambda_i^{[k]}\eta_i^{[k]}}{M}}} = \eta_i^{[k]}. \tag{62}$$

Thus, the CGAP-SBL update continuously approaches the conventional SBL update as the adaptive penalty vanishes. ■

Proposition 4 establishes that conventional SBL is recovered exactly when the spatial interaction matrix is zero. More generally, whenever the evaluated interaction for an individual MTD is non-positive, the positive-part operation sets its adaptive penalty to zero, yielding the conventional SBL update for that MTD at the corresponding iteration. The result also confirms that spatial penalization can be disabled without modifying the posterior estimation procedure.

D. Convergence Considerations

The preceding propositions establish the existence, uniqueness, and shrinkage properties of the CGAP-SBL hyperparameter update. Although these results provide an analytical characterization of the proposed penalized M-step, the convergence behavior of the complete iterative framework requires separate consideration.

At iteration k, the posterior second-order statistics $\eta_i^{[k]}$ and adaptive penalty coefficients $\lambda_i^{[k]}$ are computed using the preceding hyperparameter estimates $\boldsymbol{\gamma}^{[k-1]}$. Both quantities remain fixed during the current M-step. As established in Proposition 1, the closed-form update in (37) uniquely maximizes each scalar subproblem. Consequently, the updated hyperparameter vector satisfies

$$\mathcal{J}_k(\boldsymbol{\gamma}^{[k]}) \geq \mathcal{J}_k(\boldsymbol{\gamma}^{[k-1]}), \tag{63}$$

where $\mathcal{J}_k$ denotes the iteration-specific penalized EM objective defined in (30).

However, the penalty coefficients $\lambda_i^{[k]}$ are recomputed at successive iterations according to (29), while the posterior statistics are updated through the E-step. Therefore, the penalized objectives $\mathcal{J}_k$ and $\mathcal{J}_{k+1}$ are generally different. The improvement established in (63) applies to the current penalized M-step and does not imply monotonic improvement of a single fixed objective across iterations.

In particular, substituting (30) into (63) gives

$$\begin{aligned} &Q(\boldsymbol{\gamma}^{[k]};\boldsymbol{\gamma}^{[k-1]}) - Q(\boldsymbol{\gamma}^{[k-1]};\boldsymbol{\gamma}^{[k-1]}) \\ &\geq \sum_{i=1}^{N} \lambda_i^{[k]} \left(\gamma_i^{[k]} - \gamma_i^{[k-1]}\right). \end{aligned} \tag{64}$$

Since the right-hand side is not necessarily non-negative, an increase in the conventional EM auxiliary function, and hence monotonic improvement of the marginal likelihood, is not guaranteed by the penalized update.

Furthermore, the iteration-dependent penalization is not established as a majorization-minimization procedure for a fixed global objective. Accordingly, the uniqueness of each penalized M-step maximizer should not be interpreted as a guarantee of convergence to a global or local optimum of the complete iterative procedure.

In practice, CGAP-SBL is terminated when the relative change in the hyperparameter vector satisfies (39), or when the maximum number of iterations K is reached. The convergence behavior of the proposed framework is therefore investigated numerically in Section VIII, including the evolution of the channel estimation error and hyperparameter updates across iterations.

VI. ANC-Based Spatial Interaction Matrix Construction

A. ANC Interaction Matrix Construction

The CGAP-SBL framework developed in Section IV incorporates spatial information through a symmetric interaction matrix $\mathbf{\Omega}$. Although the framework is independent of a particular interaction matrix construction, its performance depends on how the modeled spatial relationships among MTDs are represented. To exploit the spatial dependence arising from event-driven MTD activity, an ANC approach is employed to construct the interaction matrix.

Using the distance-based spatial dependence matrix $\mathbf{C}$ defined in (11), the ANC interaction matrix is constructed as

$$\mathbf{\Omega}_{\mathrm{ANC}} = \alpha\left(\beta \mathbf{1}_{N\times N} - \mathbf{C}\right), \tag{65}$$

where $\alpha \geq 0$ and $\beta \geq 0$ are the ANC design hyperparameters.

The (i,j)-th element of the interaction matrix is therefore

$$\Omega_{ij}^{\mathrm{ANC}} = \alpha(\beta - C_{ij}). \tag{66}$$

For $\alpha > 0$, the sign of each interaction coefficient depends on the relationship between β and C_{ij}. When $C_{ij} > \beta$, the corresponding coefficient is negative, whereas $C_{ij} < \beta$ produces a positive coefficient. Thus, ANC adjusts the modeled spatial relationships relative to a common reference level determined by β.

Since $\mathbf{C}$ is symmetric, the resulting matrix $\mathbf{\Omega}_{\mathrm{ANC}}$ is also symmetric. However, it is not necessarily positive semidefinite and is not interpreted as a Gaussian precision matrix. Instead, it provides the spatial interaction information used to construct the adaptive penalty coefficients within the CGAP-SBL framework. In particular, negative entries of $\mathbf{\Omega}_{\mathrm{ANC}}$ represent interaction contributions and should not be interpreted as negative statistical correlations between binary MTD activity states.

B. ANC-Induced Adaptive Penalization

Substituting the ANC interaction matrix in (65) into the general CGAP-SBL penalty definition in (29) gives

$$\lambda_i^{[k]} = \left[\alpha \left(\beta \sum_{j=1}^{N} \gamma_j^{[k-1]} - \sum_{j=1}^{N} C_{ij} \gamma_j^{[k-1]} \right) \right]_+ . \tag{67}$$

Since $\alpha \geq 0$, this expression can equivalently be written as

$$\lambda_i^{[k]} = \alpha \left[\beta \sum_{j=1}^{N} \gamma_j^{[k-1]} - \sum_{j=1}^{N} C_{ij} \gamma_j^{[k-1]} \right]_+ . \tag{68}$$

The first term inside the positive-part operator represents a reference level determined by the aggregate hyperparameter magnitude, whereas the second term represents the spatially weighted hyperparameter support associated with the i-th MTD. Accordingly, the penalty coefficient is governed by the difference between the reference level and the modeled spatial support.

When the spatially weighted support is sufficiently large, the interaction becomes non-positive and the corresponding adaptive penalty vanishes. In this case, the hyperparameter update reduces to the conventional SBL update. Conversely, when the spatially weighted support is below the reference level, a positive penalty is applied, introducing additional shrinkage through (37).

The ANC design hyperparameters α and β serve different purposes. The parameter α controls the magnitude of the positive spatial penalties, while β determines the reference level against which the spatially weighted hyperparameter support is evaluated. Both parameters are selected externally through numerical hyperparameter tuning and remain fixed throughout each algorithm run. In contrast, the adaptive penalty coefficients $\lambda_i^{[k]}$ are recomputed at each iteration using the preceding variance hyperparameter estimates. The detailed parameter selection procedure is presented in Section VIII.

C. Properties of the ANC Spatial Interaction

The relationship between spatially weighted hyperparameter support and adaptive penalization can be established analytically.

For notational convenience, define the aggregate hyperparameter magnitude and spatially weighted support at iteration $k-1$ as

$$G^{[k-1]} = \sum_{j=1}^{N} \gamma_j^{[k-1]}, \qquad s_i^{[k-1]} = \sum_{j=1}^{N} C_{ij} \gamma_j^{[k-1]}. \tag{69}$$

The ANC-induced penalty in (68) can then be expressed as

$$\lambda_i^{[k]} = \alpha \left[\beta G^{[k-1]} - s_i^{[k-1]} \right]_+ . \tag{70}$$

Proposition 5: Consider the ANC-induced adaptive penalty in (70) with $\alpha > 0$ and fixed $G^{[k-1]} > 0$. The penalty coefficient $\lambda_i^{[k]}$ is a continuous, non-increasing function of the spatially weighted support $s_i^{[k-1]}$. Furthermore, for fixed posterior second-order statistic $\eta_i^{[k]} > 0$, the corresponding CGAP-SBL hyperparameter update is a continuous, non-decreasing function of $s_i^{[k-1]}$.

Proof: For simplicity, the iteration and device indices are omitted. From (70), the adaptive penalty is given by

$$\lambda(s) = \begin{cases} \alpha(\beta G - s), & s < \beta G, \\ 0, & s \geq \beta G. \end{cases} \tag{71}$$

Consequently, $\lambda(s)$ is continuous and non-increasing with respect to s. In the positive-penalty region,

$$\frac{\partial \lambda}{\partial s} = -\alpha < 0. \tag{72}$$

From Proposition 2, the CGAP-SBL update satisfies

$$\frac{\partial \gamma}{\partial \lambda} = -\frac{\gamma^2}{M + 2\lambda\gamma} < 0. \tag{73}$$

Applying the chain rule in the positive-penalty region gives

$$\frac{\partial \gamma}{\partial s} = \frac{\alpha\gamma^2}{M + 2\lambda\gamma} > 0. \tag{74}$$

When $s > \beta G$, the penalty vanishes and $\gamma = \eta$, so the derivative with respect to s is zero. At $s = \beta G$, the penalty and the hyperparameter update remain continuous. Therefore, for fixed posterior statistics and aggregate hyperparameter magnitude, the updated hyperparameter is a continuous, non-decreasing function of spatially weighted support. ■

Proposition 5 establishes that ANC assigns smaller penalties to MTDs with greater spatially weighted hyperparameter support, under fixed aggregate hyperparameter magnitude and posterior statistics. Consequently, the proposed method can reduce the additional shrinkage applied to MTDs that receive stronger support from the modeled spatial dependence structure. This property characterizes the behavior of the hyperparameter update but does not guarantee improved activity detection or channel estimation performance for arbitrary traffic patterns.

D. CGAP-SBL-ANC Implementation

The proposed CGAP-SBL-ANC algorithm is obtained by supplying the ANC interaction matrix in (65) to the general CGAP-SBL procedure described in Algorithm 1. The distance-based spatial dependence matrix $\mathbf{C}$ is first constructed using the MTD locations and the spatial correlation model in (11). The ANC parameters α and β are then used to determine the interaction matrix.

During each iteration, the posterior statistics are computed using the conventional SBL E-step. The ANC-induced adaptive penalties are subsequently evaluated through (68), and the hyperparameters are updated using the closed-form expression in (37). Following termination, the effective channel matrix and MTD activity states are estimated as described in Section IV.

In particular, setting $\alpha = 0$ gives $\mathbf{\Omega}_{\mathrm{ANC}} = \mathbf{0}$, and the algorithm reduces exactly to conventional SBL, as established in Proposition 4. For $\alpha > 0$, the ANC construction introduces spatially dependent penalization through the modeled relationships contained in $\mathbf{C}$.

The ANC construction can be implemented without explicitly forming the dense matrix $\mathbf{\Omega}_{\mathrm{ANC}}$. Instead, the adaptive

penalties can be computed directly from the aggregate hyperparameter magnitude and the matrix-vector product $\mathbf{C}\boldsymbol{\gamma}^{[k-1]}$. This permits efficient implementation when the spatial dependence matrix is sparse due to its finite correlation distance U. The computational requirements of the proposed framework and its ANC-based implementation are analyzed in the following section.

VII. Computational Complexity Analysis

The computational complexity of the proposed CGAP-SBL framework is analyzed in comparison with conventional SBL, with particular attention to the additional operations required for adaptive spatial penalization. The analysis considers the number of MTDs N, the number of BS antennas M, and the pilot sequence length L. Since the considered grant-free MTC system employs non-orthogonal pilot sequences with $L < N$, the posterior computations are performed using the $L \times L$ observation covariance matrix rather than directly inverting an $N \times N$ matrix.

A. Complexity of Conventional SBL

The computational cost of conventional SBL is primarily determined by the posterior estimation step. At each iteration, the observation covariance matrix $\mathbf{\Pi}$ is constructed according to (16). Since $\mathbf{\Gamma}$ is diagonal, this requires $\mathcal{O}(L^2N)$ arithmetic operations.

For $\sigma^2 > 0$ and non-negative variance hyperparameters, $\mathbf{\Pi}$ is Hermitian positive definite. Consequently, the linear systems required for posterior estimation can be solved using Cholesky factorization, avoiding explicit matrix inversion. The factorization requires $\mathcal{O}(L^3)$ operations, while solving the corresponding systems for the M received measurement vectors and the N pilot vectors requires $\mathcal{O}(L^2(M+N))$ operations.

The posterior mean in (18) can then be evaluated with an additional cost of $\mathcal{O}(LNM)$. Moreover, the full posterior covariance matrix need not be explicitly constructed because the hyperparameter update requires only its diagonal elements. Let $\boldsymbol{\theta}_i$ denote the i-th column of $\mathbf{\Theta}$. From (19), the corresponding posterior variance is obtained as

$$\Sigma_{ii} = \gamma_i - \gamma_i^2 \boldsymbol{\theta}_i^H \mathbf{\Pi}^{-1} \boldsymbol{\theta}_i. \tag{75}$$

Using the previously computed solutions involving $\mathbf{\Pi}$, all posterior diagonal elements can be evaluated with an additional cost of $\mathcal{O}(LN)$. The posterior second-order statistics and conventional SBL hyperparameter updates require $\mathcal{O}(NM)$ and $\mathcal{O}(N)$ operations, respectively.

Combining these operations, the per-iteration computational complexity of conventional SBL is

$$\mathcal{C}_{\mathrm{SBL}} = \mathcal{O}\left(L^3 + L^2(N+M) + LNM\right). \tag{76}$$

B. Complexity of CGAP-SBL

The proposed CGAP-SBL framework retains the posterior estimation procedure of conventional SBL. Therefore, its E-step has the same computational complexity as that described in the preceding subsection.

The additional computational cost arises from evaluating the adaptive penalty coefficients in (29). For a general dense interaction matrix $\mathbf{\Omega} \in \mathbb{R}^{N\times N}$, computing the matrix-vector product $\mathbf{\Omega}\boldsymbol{\gamma}^{[k-1]}$ requires $\mathcal{O}(N^2)$ operations. Applying the elementwise positive-part operator requires $\mathcal{O}(N)$ additional operations.

Once the penalty coefficients are available, the closed-form hyperparameter update in (37) is evaluated independently for each MTD, requiring $\mathcal{O}(N)$ operations. Consequently, the per-iteration complexity of CGAP-SBL with a general dense spatial interaction matrix is

$$\mathcal{C}_{\mathrm{CGAP-SBL}} = \mathcal{O}\left(L^3 + L^2(N+M) + LNM + N^2\right). \tag{77}$$

The additional $\mathcal{O}(N^2)$ term represents the computational overhead introduced by dense spatial interaction processing. However, this cost can be reduced when the interaction matrix admits a sparse or otherwise structured representation.

C. Complexity of CGAP-SBL-ANC

For CGAP-SBL-ANC, the interaction matrix follows the structure in (65). Accordingly, the adaptive penalty coefficients can be evaluated directly as

$$\lambda_i^{[k]} = \alpha \left[\beta G^{[k-1]} - s_i^{[k-1]}\right]_+, \tag{78}$$

where $G^{[k-1]}$ and $s_i^{[k-1]}$ are defined in (69).

The aggregate hyperparameter magnitude $G^{[k-1]}$ can be computed in $\mathcal{O}(N)$ operations. The spatially weighted support vector is obtained from the matrix-vector product $\mathbf{C}\boldsymbol{\gamma}^{[k-1]}$. When $\mathbf{C}$ is stored as a dense matrix, this operation requires $\mathcal{O}(N^2)$ operations, resulting in the same asymptotic per-iteration complexity as general dense CGAP-SBL.

However, the spatial dependence model in (11) assigns zero coefficients to MTD pairs separated by more than the correlation cutoff distance U. This permits sparse storage and multiplication when the resulting matrix contains relatively few nonzero entries. Denoting the number of nonzero elements of $\mathbf{C}$ by $\mathrm{nnz}(\mathbf{C})$, the spatially weighted support can be evaluated in $\mathcal{O}(\mathrm{nnz}(\mathbf{C}))$ operations.

Therefore, the per-iteration computational complexity of CGAP-SBL-ANC with a sparse implementation is

$$\mathcal{C}_{\mathrm{CGAP-SBL-ANC}} = \mathcal{O}\left(L^3 + L^2(N+M) + LNM + \mathrm{nnz}(\mathbf{C}) + N\right). \tag{79}$$

This implementation avoids explicitly forming the dense matrix $\mathbf{\Omega}_{\mathrm{ANC}}$. Nevertheless, the sparsity advantage depends on the device geometry and correlation cutoff distance. If $\mathbf{C}$ is dense, then $\mathrm{nnz}(\mathbf{C}) = \mathcal{O}(N^2)$, and the interaction cost remains quadratic in N.

The preceding expressions characterize the computational cost per iteration, excluding the one-time construction of the spatial dependence matrix. Direct evaluation of all pairwise MTD distances requires $\mathcal{O}(N^2)$ operations during initialization. If $\mathbf{C}$ is constructed once and reused throughout the iterations, this preprocessing cost is not incurred at every iteration.

The overall computational requirements also depend on the number of iterations performed before termination. Accordingly, numerical comparisons of execution time and convergence behavior are presented in Section VIII to complement the asymptotic complexity analysis.

VIII. Numerical Results and Discussion

Numerical simulations are considered to evaluate the proposed CGAP-SBL-ANC algorithm in terms of MTD activity detection accuracy, channel estimation performance, and convergence behavior. The proposed method is compared with conventional SBL, OMP, and CoSaMP under event-driven spatially correlated MTD activity. A separate experiment investigates the influence of spatial activity dependence by comparing event-driven and spatially randomized activity patterns while preserving the realized number of active MTDs.

A. Simulation Setup and Performance Metrics

A single-cell MTC network is considered, where a BS equipped with $M = 4$ antennas serves $N = 400$ single-antenna MTDs uniformly distributed within a circular coverage area of radius $R = 40$ meters. The spatially correlated MTD activity patterns are generated according to the event-driven model described in Section III, with $V = 2$ events, an influence distance of $D = 15$ meters, and a spatial decay parameter of $\kappa = 3.78$. The decay parameter is selected to obtain an average of approximately 10 active MTDs per realization. Consequently, the actual number of active MTDs varies across realizations. The spatial dependence matrix $\mathbf{C}$ is constructed using the same spatial parameters as the event-driven activity model, i.e., $U = D = 15$ meters and $\rho = \kappa = 3.78$ meters.

Each MTD is assigned a unit-norm non-orthogonal pilot sequence consisting of independently generated QPSK symbols. The pilot length L is varied according to the experimental configuration, as specified in the corresponding performance evaluations. The channels follow the block Rayleigh fading model described in Section II, with normalized large-scale fading coefficients $g_i = 1$. The simulation results are averaged over $Q = 1000$ independent Monte Carlo realizations, indexed by $q = 1, \ldots, Q$. A separate set of 100 realizations is used for hyperparameter tuning.

The proposed CGAP-SBL-ANC algorithm is compared with conventional SBL [11], CoSaMP [38], and OMP [39]. The OMP and CoSaMP implementations employ the multiple-measurement-vector (MMV) formulation, jointly recovering the active support across the BS antennas. For each realization, all evaluated algorithms use the same received signal matrix $\mathbf{Y}$ and pilot matrix $\mathbf{\Theta}$. The true effective channel matrix $\mathbf{Z}$ is used only for performance evaluation and is not supplied to the recovery algorithms.

For the OMP and CoSaMP benchmarks, let S_q denote the true number of active MTDs in realization q. The true active-device count is provided as oracle sparsity information, while the identities of the active MTDs remain unknown to both algorithms. Their sparsity budget is defined as

$$B_q = \min(S_q, L, N), \tag{80}$$

where L denotes the pilot length of the corresponding experimental configuration. Thus, the budget adapts to both the realized activity level and the pilot length. In contrast, conventional SBL and CGAP-SBL-ANC do not require prior knowledge of S_q. Among the $Q = 1000$ evaluation realizations generated by the event-driven model, only two contain more than 20 active MTDs. Consequently, sparsity-budget truncation occurs only for these two realizations when $L = 20$, while their true activity patterns remain unchanged during performance evaluation.

In the MMV-OMP implementation, active-device candidates are selected sequentially according to the energy of the correlations between the pilot sequences and the current residual across the BS antennas. After each selection, the channel coefficients associated with the selected support are re-estimated using least squares, and the residual is updated. The algorithm performs B_q support-selection steps.

In the MMV-CoSaMP implementation, candidate support elements are identified using residual correlations and merged with the current support. Channel coefficients are estimated over the merged support using least squares, after which the estimate is pruned to B_q support elements according to the recovered row energies. A least-squares refinement is subsequently performed over the retained support, and the residual is updated. The procedure continues until the stopping criterion is satisfied or the maximum number of CoSaMP iterations is reached.

For both OMP and CoSaMP, the final effective channel estimates are obtained from the least-squares solutions over their respective estimated supports, with all remaining rows set to zero.

Channel estimation accuracy is quantified using the normalized mean squared error (NMSE), defined as

$$\mathrm{NMSE} = \mathbb{E}\left[\frac{\left\|\mathbf{Z} - \widehat{\mathbf{Z}}\right\|_F^2}{\|\mathbf{Z}\|_F^2}\right], \tag{81}$$

where $\mathbf{Z}$ and $\widehat{\mathbf{Z}}$ denote the true and estimated effective channel matrices, respectively [21]. The NMSE is computed separately for each Monte Carlo realization and averaged over realizations with nonzero true channel energy, for which the normalization is well defined.

For the SBL-based methods, the effective channel estimates are obtained from the final posterior mean in (40), without applying the binary activity-detection threshold to the estimated channel coefficients. For OMP and CoSaMP, the corresponding estimates are obtained from their final sparse least-squares recovery solutions. Thus, all methods are evaluated using the same NMSE definition, although their channel estimation procedures differ.

MTD activity detection accuracy is evaluated using the F1 score, which balances precision and recall:

$$\mathrm{F1} = 2\,\frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{82}$$

The corresponding precision and recall are defined as

$$\text{Precision} = \frac{|\mathcal{S} \cap \widehat{\mathcal{S}}|}{|\widehat{\mathcal{S}}|}, \tag{83}$$

$$\text{Recall} = \frac{|\mathcal{S} \cap \widehat{\mathcal{S}}|}{|\mathcal{S}|}, \tag{84}$$

where $\mathcal{S}$ and $\widehat{\mathcal{S}}$ denote the true and estimated active supports, respectively. For each realization, the true active support is

$$\mathcal{S} = \{i : a_i = 1\}. \tag{85}$$

For CGAP-SBL-ANC and conventional SBL, the estimated active support is obtained by thresholding the final variance hyperparameters:

$$\widehat{\mathcal{S}}^{(A)} = \left\{ i : \widehat{\gamma}_i^{(A)} \geq \tau_A \right\}, \tag{86}$$

where $A \in \{\text{CGAP}, \text{SBL}\}$ identifies the corresponding method, $\widehat{\gamma}_i^{(A)}$ is its final estimated variance hyperparameter, and τ_A is its activity-detection threshold. The detection rule for the proposed framework is given in (41). The thresholds for CGAP-SBL-ANC and conventional SBL are selected separately using the tuning realizations.

For OMP and CoSaMP, the estimated active support is obtained from the energies of the estimated effective channel vectors:

$$\widehat{\mathcal{S}}^{(A,q)} = \text{Top}_{B_q} \left(\left\{ \left\| \widehat{\mathbf{z}}_i^{(A,q)} \right\|_2^2 \right\}_{i=1}^{N} \right), \tag{87}$$

where $A \in \{\text{OMP}, \text{CoSaMP}\}$, $\widehat{\mathbf{z}}_i^{(A,q)} \in \mathbb{C}^M$ denotes the estimated effective channel vector of the i-th MTD in realization q, and $\text{Top}_{B_q}(\cdot)$ returns the indices corresponding to the B_q largest channel-vector energies. Therefore, unlike the SBL-based methods, OMP and CoSaMP do not employ variance-hyperparameter thresholding for activity detection.

For all methods, the F1 score is calculated from the corresponding estimated support and averaged across Monte Carlo realizations. When a precision or recall denominator is zero, the corresponding value is set to zero. The F1 score reaches its maximum value of one when the estimated support exactly matches a nonempty true active support [21], [40].

B. ANC Design Hyperparameter Selection

The CGAP-SBL-ANC algorithm employs ANC design hyperparameters α and β, which determine the strength and reference level of the adaptive spatial penalties, respectively, as described in Section VI. These design hyperparameters are selected through numerical tuning and remain fixed during subsequent algorithm iterations. In contrast, the device-specific penalty coefficients are updated adaptively according to the evolving variance hyperparameters.

The maximum number of algorithm iterations is set to $K = 500$, and the convergence tolerance is set to $\delta = 10^{-4}$, according to the stopping criterion in (39). Activity-detection thresholds for CGAP-SBL-ANC and conventional SBL are selected independently using the tuning realizations.

To investigate the influence of the ANC design hyperparameters, a grid search is considered over $\alpha \in \{2, 4, 6, 8, 10, 12, 14\}$ and $\beta \in \{0.05, 0.10, \ldots, 0.35\}$. The evaluation is performed at an SNR of $12\,\text{dB}$ and pilot length $L = 25$, using 100 independent Monte Carlo realizations.

Fig. 2 is reserved for the corresponding F1 and NMSE contours over the considered hyperparameter space. These evaluations are intended to characterize the sensitivity of the proposed method to the ANC design parameters and identify a suitable operating configuration. The selected values are subsequently kept fixed for the main evaluation experiments.

C. MTD Activity Detection Performance

MTD activity detection performance is evaluated using the F1 score under event-driven spatially correlated activity. Two experimental settings are considered: (i) a fixed SNR of $12\,\text{dB}$ with varying pilot lengths $L \in \{20, 25, 30, 35, 40, 45\}$, and (ii) a fixed pilot length $L = 30$ with varying SNR values in $\{4, 8, 12, 16, 20, 24\}\,\text{dB}$.

The corresponding performance comparisons are presented in Fig. 3, where CGAP-SBL-ANC is evaluated against conventional SBL, CoSaMP, and OMP using identical received observations within each realization.

The pilot-length experiment examines the influence of the available observation dimension on active-support recovery. Short pilot sequences provide fewer observations relative to the number of potential MTDs, making activity detection more challenging. The SNR experiment separately investigates the influence of measurement noise on support recovery.

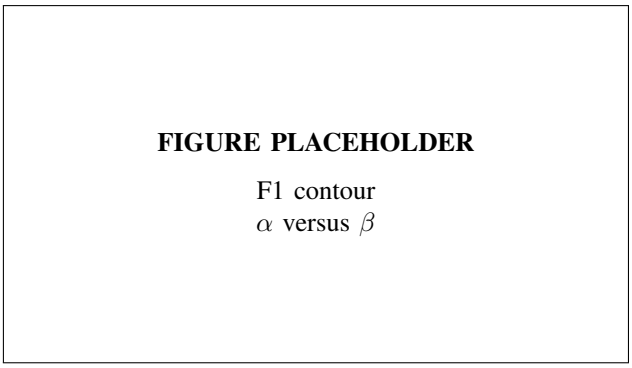
The comparison is intended to assess whether the adaptive correlation-guided penalization mechanism provides an advantage over conventional SBL, which does not incorporate spatial interaction information, and over the MMV greedy recovery methods, which use oracle sparsity budgets. In particular, the F1 curves allow the effect of spatially guided variance-hyperparameter updates to be evaluated across different pilot-resource and noise conditions.

D. Channel Estimation Performance

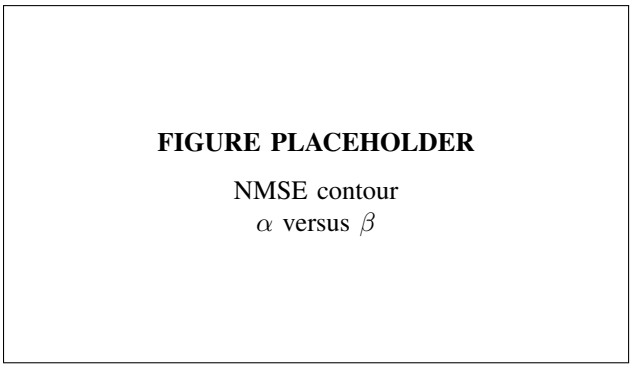
Channel estimation performance is evaluated using NMSE under the same pilot-length and SNR configurations considered for activity detection. The corresponding comparisons between CGAP-SBL-ANC, conventional SBL, CoSaMP, and OMP are presented in Fig. 4.

The pilot-length experiment investigates the effect of the available pilot observations on effective channel recovery, while the SNR experiment examines the sensitivity of channel estimation to measurement noise. All algorithms estimate the effective channel matrix from the same pilot observations within each realization, and their estimates are evaluated against the corresponding true matrix $\mathbf{Z}$.

For CGAP-SBL-ANC and conventional SBL, channel estimation is performed using the final posterior mean without applying the binary activity-detection mask. Consequently, their NMSE values reflect the accuracy of the continuous effective channel estimates rather than the result of hard activity decisions.

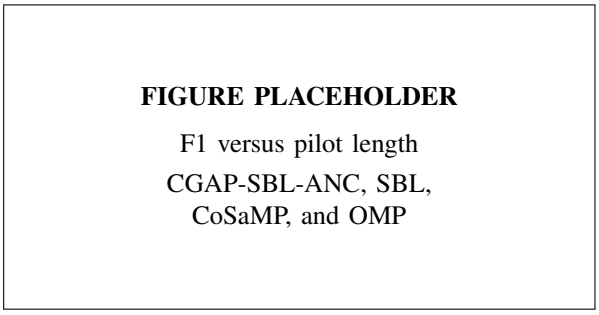


(a) F1 score over the α–β hyperparameter space.

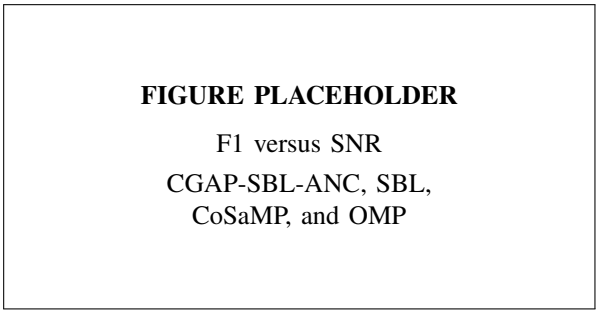


(b) NMSE over the α–β hyperparameter space.

Fig. 2. Sensitivity of CGAP-SBL-ANC to the ANC design hyperparameters α and β: (a) F1 score and (b) NMSE, evaluated at an SNR of $12\,\mathrm{dB}$ and pilot length $L = 25$.

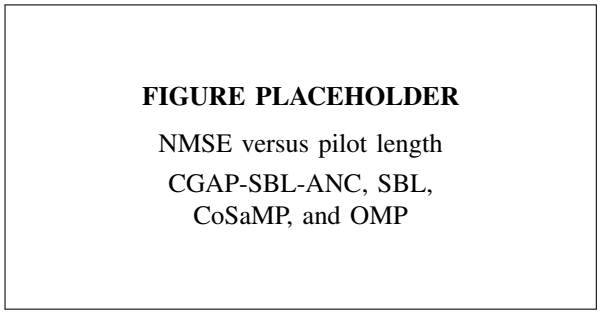


(a) F1 score versus pilot length at fixed SNR of $12\,\mathrm{dB}$.

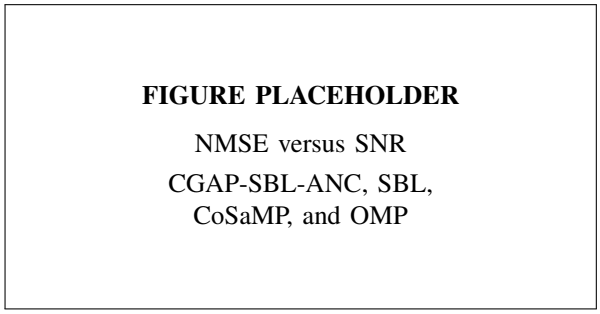


(b) F1 score versus SNR at fixed pilot length $L = 30$.

Fig. 3. MTD activity detection performance under event-driven spatially correlated activity, comparing CGAP-SBL-ANC with conventional SBL, CoSaMP, and OMP.



(a) NMSE versus pilot length at fixed SNR of $12\,\mathrm{dB}$.



(b) NMSE versus SNR at fixed pilot length $L = 30$.

Fig. 4. Channel estimation performance under event-driven spatially correlated activity, comparing CGAP-SBL-ANC with conventional SBL, CoSaMP, and OMP.

In contrast, OMP and CoSaMP reconstruct sparse channel matrices through support selection and least-squares estimation. Their estimated channel coefficients are restricted to the selected support, and all remaining coefficients are set to zero.

The ANC-induced penalties affect channel estimation indirectly through the variance-hyperparameter updates and the resulting posterior distribution. When the modeled spatial information is informative, these penalties may suppress unsupported channel components while limiting additional shrinkage for components receiving stronger spatial support. This mechanism is consistent with the adaptive shrinkage properties established in Proposition 2.

The NMSE comparisons are therefore intended to quantify the effect of incorporating spatial interaction information into the Bayesian recovery procedure relative to conventional SBL and the considered greedy recovery benchmarks.

E. Convergence Behavior

The iterative behavior of CGAP-SBL-ANC is investigated by examining the evolution of channel estimation NMSE over a maximum of $K = 500$ iterations at a fixed SNR of $12\,\mathrm{dB}$ and pilot length $L = 30$. The convergence behavior is compared with conventional SBL under the same received observations and initialization settings.

Fig. 5 is reserved for the corresponding NMSE trajectories. The comparison is intended to examine how the adaptive spatial penalties influence the evolution of the channel estimates relative to conventional SBL.

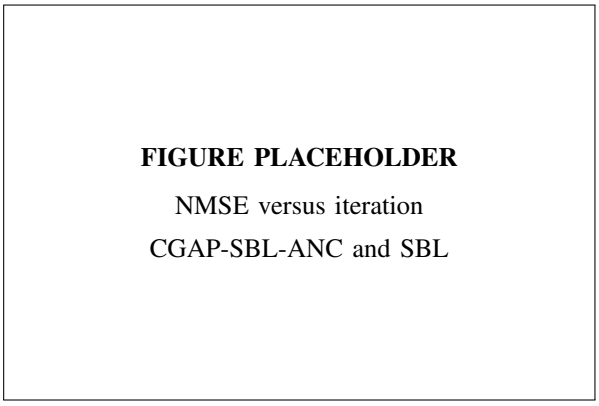


Fig. 5. Evolution of channel estimation NMSE over iterations at an SNR of 12 dB and pilot length $L = 30$, comparing CGAP-SBL-ANC with conventional SBL under event-driven spatially correlated activity.

Both methods update their variance hyperparameters iteratively using posterior second-order statistics. The proposed method additionally incorporates spatially informed penalties into the hyperparameter updates, whereas conventional SBL performs the corresponding updates without the spatial interaction term.

The NMSE trajectories allow the numerical evolution of the estimated effective channel matrix to be examined across successive iterations. In particular, the comparison can reveal differences in the initial rate of error reduction, the number of iterations required to reach a stable estimate, and the final estimation-error level.

The convergence behavior observed in numerical simulations should not be interpreted as a mathematical guarantee of convergence of the complete CGAP-SBL algorithm. As discussed in Section V, the theoretical results establish the unique optimizer of each iteration-specific penalized M-step but do not guarantee monotonic improvement of a fixed global objective across all iterations.

F. Impact of Spatial Activity Dependence

To investigate the contribution of spatial activity dependence to the proposed CGAP-SBL-ANC algorithm, a controlled comparison is considered between event-driven correlated activity and spatially randomized activity. The purpose of this experiment is to examine whether the performance advantage associated with correlation-guided penalization depends on the relationship between MTD activity and the spatial information used to construct the ANC interaction matrix.

For each Monte Carlo realization q, the event-driven activity model generates a binary activity vector $\mathbf{a}^{(\mathrm{corr},q)}$ containing S_q active MTDs. A corresponding spatially randomized activity vector $\mathbf{a}^{(\mathrm{rand},q)}$ is generated by uniformly selecting S_q distinct MTDs without replacement from the same set of N devices.

Consequently, the two activity vectors satisfy

$$\left\|\mathbf{a}^{(\mathrm{corr},q)}\right\|_0 = \left\|\mathbf{a}^{(\mathrm{rand},q)}\right\|_0 = S_q. \tag{88}$$

In the correlated scenario, MTD activation is governed by the event-driven probability model described in Section III. In the randomized scenario, active-device identities are selected independently of the device locations and event positions, conditional on the realized active-device count. This procedure preserves the sparsity level of each realization while removing the event-driven spatial preference from the selection of active MTDs.

The device locations, event locations, pilot sequences, and per-device channel realizations are retained across the paired scenarios. Only the activity support is changed. Accordingly, the effective channel matrices are constructed as

$$\mathbf{Z}^{(\mathrm{corr},q)} = \mathrm{diag}\left(\mathbf{a}^{(\mathrm{corr},q)}\right)\left(\mathbf{H}^{(q)}\right)^T, \tag{89}$$

$$\mathbf{Z}^{(\mathrm{rand},q)} = \mathrm{diag}\left(\mathbf{a}^{(\mathrm{rand},q)}\right)\left(\mathbf{H}^{(q)}\right)^T. \tag{90}$$

The corresponding received signal matrices are regenerated according to

$$\mathbf{Y}^{(\mathrm{corr},q)} = \mathbf{\Theta}^{(q)}\mathbf{Z}^{(\mathrm{corr},q)} + \mathbf{W}^{(\mathrm{corr},q)}, \tag{91}$$

$$\mathbf{Y}^{(\mathrm{rand},q)} = \mathbf{\Theta}^{(q)}\mathbf{Z}^{(\mathrm{rand},q)} + \mathbf{W}^{(\mathrm{rand},q)}, \tag{92}$$

where the noise realizations are generated to maintain the prescribed SNR in each scenario. The same underlying standardized noise realization may be employed for the paired scenarios, with appropriate scaling when their received signal powers differ.

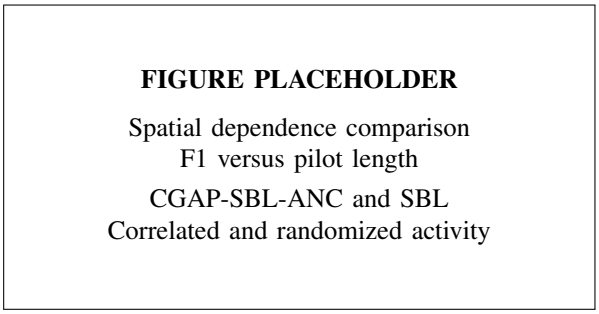
The spatial dependence matrix $\mathbf{C}$ is constructed using the original device locations and is kept unchanged between the two scenarios. Therefore, the difference between the scenarios arises from the activity-generation mechanism rather than from changes to the device geometry or the spatial interaction matrix.

Both CGAP-SBL-ANC and conventional SBL are evaluated separately under correlated and spatially randomized activity. For a given realization and activity scenario, the two algorithms receive identical pilot and received signal matrices. Their estimated supports and effective channel matrices are evaluated against the true activity vector and effective channel matrix of that scenario.

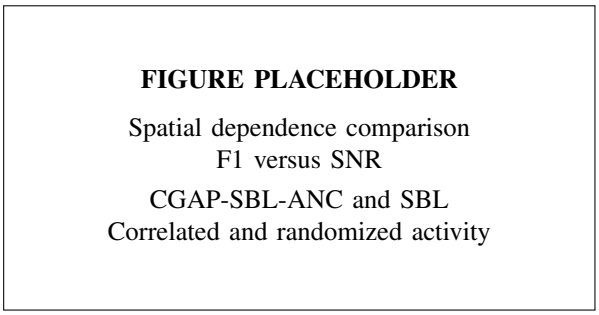
The ANC design hyperparameters and the activity-detection thresholds are kept fixed across the two activity scenarios after selection using the correlated-activity tuning set. This avoids introducing additional differences through scenario-specific parameter tuning.

Since changing the active-device support also changes the underlying received observations, conventional SBL may exhibit different performance under the two scenarios despite not using spatial information. Therefore, the contribution of correlation-guided penalization is examined by comparing the performance difference between CGAP-SBL-ANC and conventional SBL within each activity scenario.

In particular, the comparison considers the difference in F1 score and the reduction in NMSE achieved by CGAP-SBL-ANC relative to conventional SBL under correlated and spatially randomized activity. This allows the effect of informative spatial activity structure to be distinguished from performance changes that arise simply because a different set of MTDs is active.

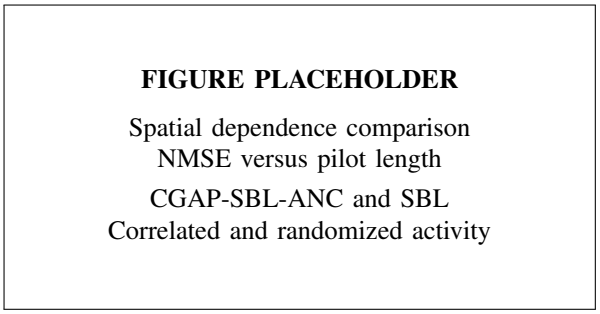


(a) F1 score versus pilot length.

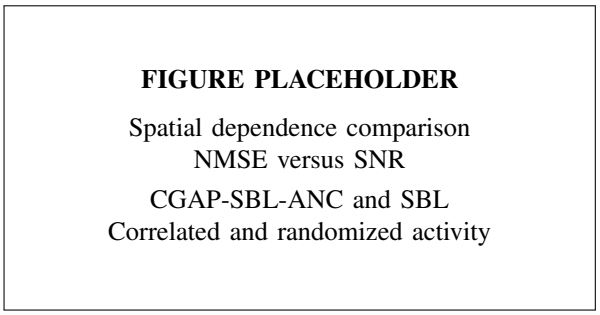


(b) F1 score versus SNR.

Fig. 6. Effect of spatial activity dependence on MTD activity detection performance: F1 score versus (a) pilot length and (b) SNR for CGAP-SBL-ANC and conventional SBL under event-driven correlated and spatially randomized activity.



(a) NMSE versus pilot length.



(b) NMSE versus SNR.

Fig. 7. Effect of spatial activity dependence on channel estimation performance: NMSE versus (a) pilot length and (b) SNR for CGAP-SBL-ANC and conventional SBL under event-driven correlated and spatially randomized activity.

1) MTD Activity Detection Under Spatial Randomization: The influence of spatial activity dependence on MTD activity detection is assessed by comparing the F1 scores of CGAP-SBL-ANC and conventional SBL under the two activity scenarios.

The evaluation considers variations in pilot length at a fixed SNR of $12\,\mathrm{dB}$ and variations in SNR at a fixed pilot length $L = 30$. The corresponding comparisons are reserved in Fig. 6.

For each experimental condition, four performance curves are considered: CGAP-SBL-ANC under correlated activity, conventional SBL under correlated activity, CGAP-SBL-ANC under spatially randomized activity, and conventional SBL under spatially randomized activity.

The comparisons are intended to determine whether correlation-guided penalization provides a larger activity-detection advantage when the active MTDs follow the event-driven spatial structure. The randomized scenario additionally examines the behavior of the proposed method when the spatial interaction matrix is no longer informative about the event-driven selection of active-device identities.

2) Channel Estimation Under Spatial Randomization: The influence of spatial activity dependence on channel estimation is evaluated using NMSE under the same pilot-length and SNR configurations. The corresponding comparisons are reserved in Fig. 7.

Both CGAP-SBL-ANC and conventional SBL use their unmasked final posterior means to estimate the effective channel matrices. For each scenario, the NMSE is calculated using the corresponding true effective channel matrix, ensuring that the channel recovery errors are measured relative to the actual active-device support.

The comparison examines whether the relative channel estimation performance of CGAP-SBL-ANC changes when the active-device support is selected independently of the spatial event structure. Particular attention is given to the NMSE difference between the proposed method and conventional SBL within each scenario, rather than solely to changes in the absolute NMSE of the proposed method.

This experiment provides a complementary evaluation of the dependence of correlation-guided penalization on the informativeness of the modeled spatial interaction structure. Any observed improvements or degradation will be interpreted relative to conventional SBL under the corresponding activity scenario.

IX. Conclusion

This paper proposed a CGAP-SBL framework for the JADCE problem in grant-free MTC systems. The true MTD activity is represented by a binary activity vector, while nonnegative continuous SBL variance hyperparameters are used to infer the corresponding sparse support. Spatial dependence is incorporated through a Gaussian-inspired quadratic interaction model parameterized by a symmetric spatial interaction matrix. Rather than interpreting this interaction as a Gaussian hyperprior, the proposed framework uses it to construct adaptive nonnegative penalties within the EM procedure, without requiring the interaction matrix to be positive semidefinite.

The CGAP-SBL framework admits a closed-form hyperparameter update that incorporates spatially informed shrinkage. Theoretical analysis establishes the existence and uniqueness of the global maximizer of each iteration-specific penalized

M-step, characterizes the resulting shrinkage and sensitivity properties, and demonstrates that the proposed update reduces to conventional SBL when the spatial interaction is removed. The ANC construction is further introduced to obtain the spatial interaction matrix from modeled pairwise spatial dependence among MTDs. The resulting implementation, termed CGAP-SBL-ANC, employs two design hyperparameters, α and β, to regulate the strength and reference level of the adaptive spatial penalties. By exploiting the structure of the ANC interaction matrix, the proposed implementation retains a computational profile comparable to conventional SBL, with limited additional overhead for spatial interaction calculations.

The numerical evaluation is designed to assess the activity detection accuracy, channel estimation performance, and empirical convergence behavior of CGAP-SBL-ANC relative to conventional SBL, OMP, and CoSaMP under event-driven spatially correlated activity. A separate controlled comparison between correlated and spatially randomized activity is also considered to investigate the contribution of informative spatial dependence. Together, these evaluations are intended to characterize the benefits and limitations of correlation-guided adaptive penalization for JADCE in dense, event-driven MTC networks.

X. Acknowledgments

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